

# PC SERVICING AND TROUBLESHOOTING (305)

**STATE – 2019**

*TOTAL POINTS* \_\_\_\_\_ *(500 points)*

**Failure to adhere to any of the following rules will result in disqualification:**

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

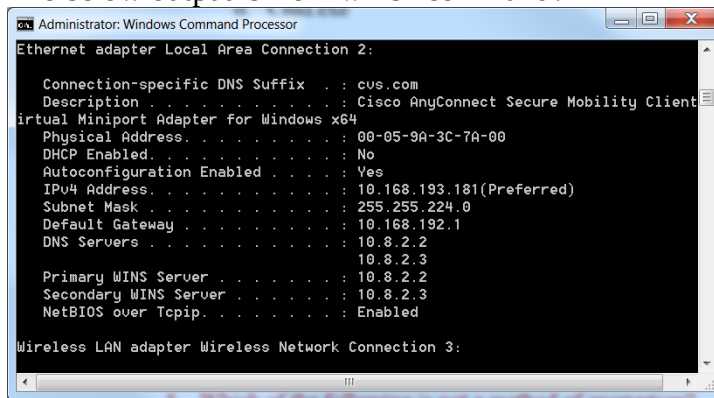
No more than sixty (60) minutes testing time

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*Workplace Skills Assessment Program* competition.

### MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question. Mark “A” if the statement is True. Mark “B” if the statement is False.

1. Command line tools are run from the command prompt. To get to the command prompt, click on start and type any of the below *except* \_\_\_\_\_ in the Search box then Enter.
  - a. command.exe
  - b. command
  - c. cmd
  - d. cmd.exe
2. The below output is from which command?

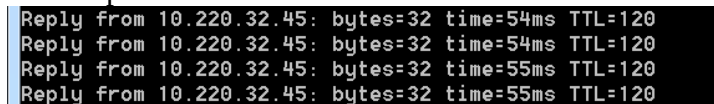


```
Administrator: Windows Command Processor
Ethernet adapter Local Area Connection 2:

    Connection-specific DNS Suffix  . : cus.com
    Description . . . . . : Cisco AnyConnect Secure Mobility Client
    Physical Address. . . . . : 00-05-9A-3C-7A-00
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 10.168.193.181(Preferred)
    Subnet Mask . . . . . : 255.255.224.0
    Default Gateway . . . . . : 10.168.192.1
    DNS Servers . . . . . : 10.8.2.2
                           10.8.2.3
    Primary WINS Server . . . . . : 10.8.2.2
    Secondary WINS Server . . . . . : 10.8.2.3
    NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wireless Network Connection 3:
```

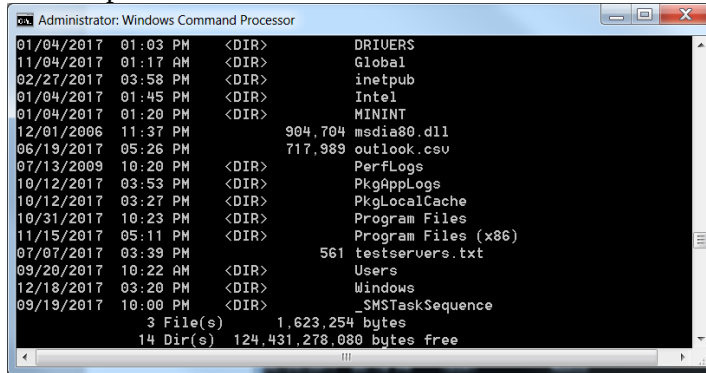
- a. ping
  - b. tracert
  - c. ipconfig
  - d. ipconfig /all
3. The output below is from which command?



```
Reply from 10.220.32.45: bytes=32 time=54ms TTL=120
Reply from 10.220.32.45: bytes=32 time=54ms TTL=120
Reply from 10.220.32.45: bytes=32 time=55ms TTL=120
Reply from 10.220.32.45: bytes=32 time=55ms TTL=120
```

- a. ipconfig
- b. ping
- c. ipconfig /all
- d. tracert

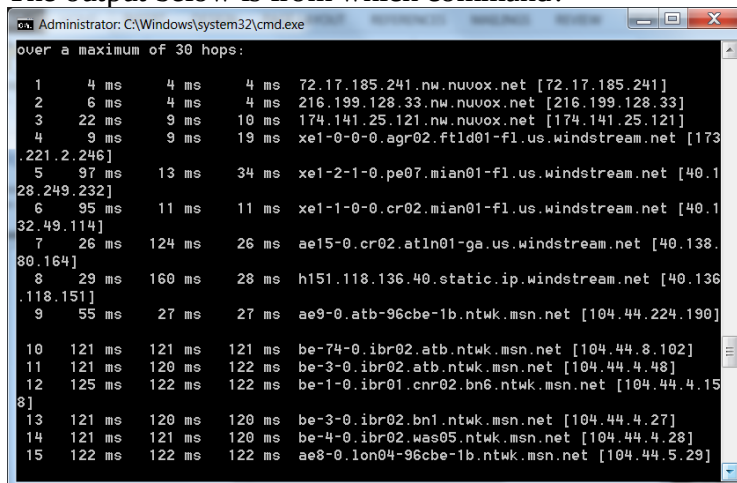
4. The output below is from which command?



```
Administrator: Windows Command Processor
01/04/2017 01:03 PM <DIR> DRIVERS
11/04/2017 01:17 AM <DIR> Global
02/27/2017 03:58 PM <DIR> inetpub
01/04/2017 01:45 PM <DIR> Intel
01/04/2017 01:20 PM <DIR> MININT
12/01/2006 11:37 PM 904,704 msdia80.dll
06/19/2017 05:26 PM 717,989 outlook.csv
07/13/2009 10:20 PM <DIR> PerfLogs
10/12/2017 03:53 PM <DIR> PkgAppLogs
10/12/2017 03:27 PM <DIR> PkgLocalCache
10/31/2017 10:23 PM <DIR> Program Files
11/15/2017 05:11 PM <DIR> Program Files (x86)
07/07/2017 03:39 PM 561 testservers.txt
09/20/2017 10:22 AM <DIR> Users
12/18/2017 03:20 PM <DIR> Windows
09/19/2017 10:00 PM <DIR> _SMSTaskSequence
3 File(s) 1,623,254 bytes
14 Dir(s) 124,431,278,080 bytes free
```

- a. dir
- b. gpresult
- c. ipcxonfig /all
- d. ping

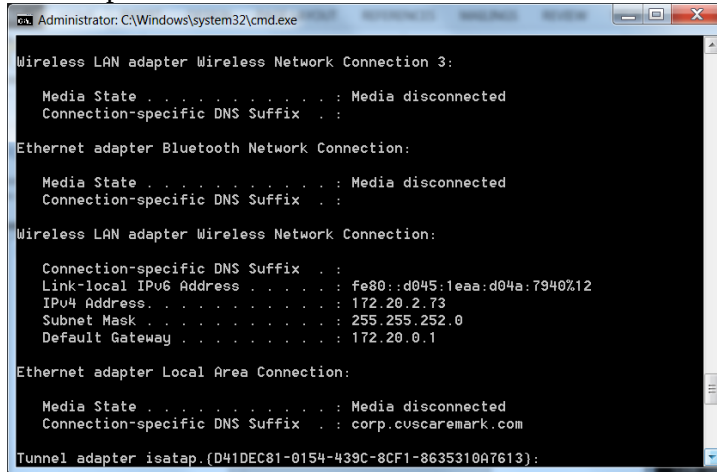
5. The output below is from which command?



```
Administrator: C:\Windows\system32\cmd.exe
over a maximum of 30 hops:
 1  4 ms  4 ms  4 ms  72.17.185.241.nw.nuvox.net [72.17.185.241]
 2  6 ms  4 ms  4 ms  216.199.128.33.nw.nuvox.net [216.199.128.33]
 3  22 ms  9 ms  10 ms  174.141.25.121.nw.nuvox.net [174.141.25.121]
 4  9 ms  9 ms  19 ms  xe1-0-0-0.agr02.ftld01-f1.us.windstream.net [173.221.2.246]
 5  97 ms  13 ms  34 ms  xe1-2-1-0.pe07.mian01-f1.us.windstream.net [40.128.249.232]
 6  95 ms  11 ms  11 ms  xe1-1-0-0.cr02.mian01-f1.us.windstream.net [40.132.49.114]
 7  26 ms  124 ms  26 ms  ae15-0.cr02.atln01-ga.us.windstream.net [40.138.80.164]
 8  29 ms  160 ms  28 ms  h151.118.136.40.static.ip.windstream.net [40.136.118.151]
 9  55 ms  27 ms  27 ms  ae9-0.atb-96cbe-1b.ntwk.msn.net [104.44.224.190]
10 121 ms 121 ms 121 ms  be-74-0.ibr02.atb.ntwk.msn.net [104.44.8.102]
11 121 ms 120 ms 122 ms  be-3-0.ibr02.atb.ntwk.msn.net [104.44.4.48]
12 125 ms 122 ms 122 ms  be-1-0.ibr01.cnr02.bn6.ntwk.msn.net [104.44.4.158]
13 121 ms 120 ms 120 ms  be-3-0.ibr02.bn1.ntwk.msn.net [104.44.4.27]
14 121 ms 121 ms 120 ms  be-4-0.ibr02.was05.ntwk.msn.net [104.44.4.28]
15 122 ms 122 ms 122 ms  ae8-0.lon04-96cbe-1b.ntwk.msn.net [104.44.5.29]
```

- a. dir
- b. ping
- c. ipconfig /all
- d. tracert

6. The output below is from which command?



```
Administrator: C:\Windows\system32\cmd.exe

Wireless LAN adapter Wireless Network Connection 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wireless Network Connection:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::d045:1eaa:d04a:7940%12
    IPv4 Address. . . . . : 172.20.2.73
    Subnet Mask . . . . . : 255.255.252.0
    Default Gateway . . . . . : 172.20.0.1

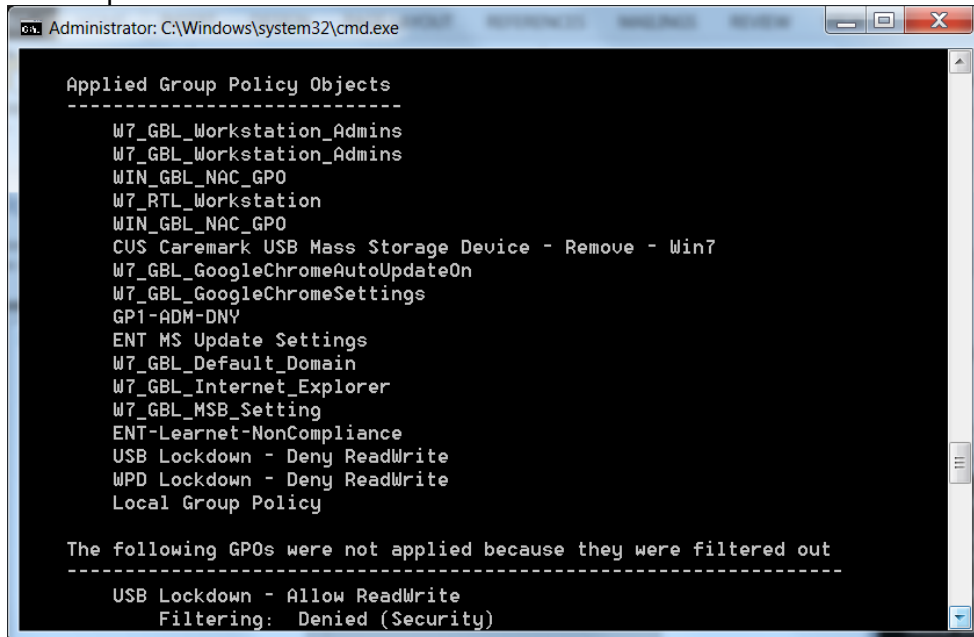
Ethernet adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : corp.cuscaremark.com

Tunnel adapter isatap.{D41DEC81-0154-439C-8CF1-8635310A7613}:
```

- a. echo
- b. dir
- c. ipconfig
- d. ping

7. The output below is from which command?



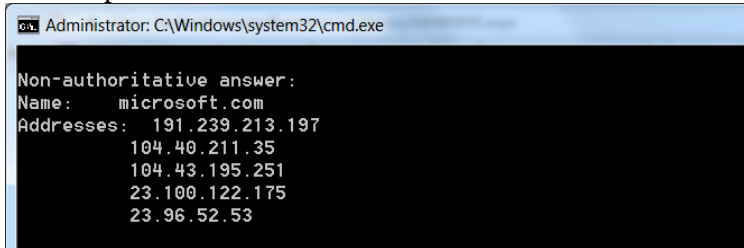
```
Administrator: C:\Windows\system32\cmd.exe

Applied Group Policy Objects
-----
W7_GBL_Workstation_Admins
W7_GBL_Workstation_Admins
WIN_GBL_NAC_GPO
W7_RTL_Workstation
WIN_GBL_NAC_GPO
CUS Caremark USB Mass Storage Device - Remove - Win7
W7_GBL_GoogleChromeAutoUpdateOn
W7_GBL_GoogleChromeSettings
GP1-ADM-DNY
ENT MS Update Settings
W7_GBL_Default_Domain
W7_GBL_Internet_Explorer
W7_GBL_MSB_Setting
ENT-Learnnet-NonCompliance
USB Lockdown - Deny ReadWrite
WPD Lockdown - Deny ReadWrite
Local Group Policy

The following GPOs were not applied because they were filtered out
-----
USB Lockdown - Allow ReadWrite
Filtering: Denied (Security)
```

- a. gpo
- b. gpresult
- c. ipconfig /all
- d. ping

8. The output below is from which command?

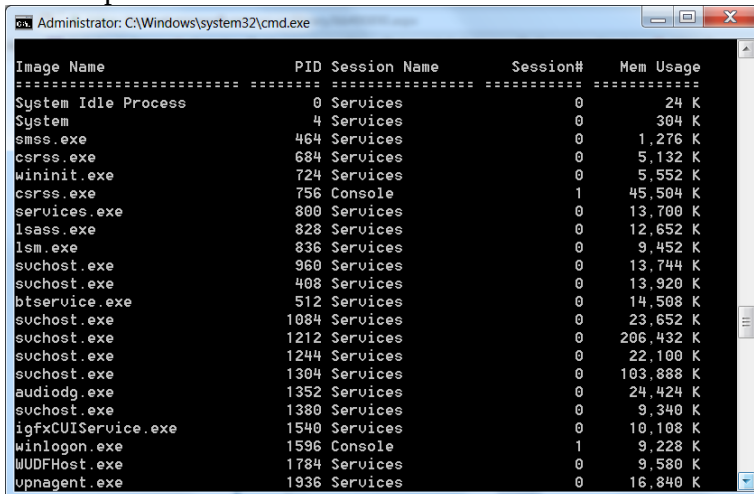


```
Administrator: C:\Windows\system32\cmd.exe

Non-authoritative answer:
Name:    microsoft.com
Addresses: 191.239.213.197
          104.40.211.35
          104.43.195.251
          23.100.122.175
          23.96.52.53
```

- a. netstat
- b. fdisk
- c. nslookup
- d. tracert

9. The output below is from which command?

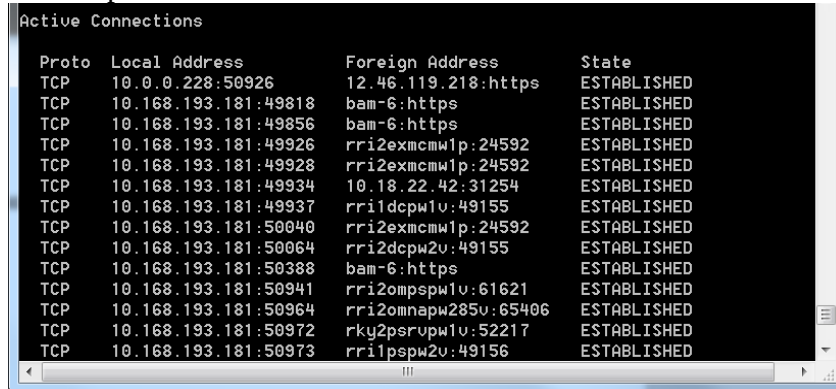


```
Administrator: C:\Windows\system32\cmd.exe

Image Name                   PID Session Name        Session#    Mem Usage
-----
System Idle Process          0 Services             0             24 K
System                        4 Services             0            304 K
smss.exe                     464 Services             0           1,276 K
csrss.exe                     684 Services             0           5,132 K
wininit.exe                   724 Services             0           5,552 K
csrss.exe                     756 Console              1          45,504 K
services.exe                  800 Services             0          13,700 K
lsass.exe                     828 Services             0          12,652 K
lsm.exe                       836 Services             0           9,452 K
svchost.exe                   960 Services             0          13,744 K
svchost.exe                   408 Services             0          13,920 K
btsservice.exe               512 Services             0          14,508 K
svchost.exe                  1084 Services             0          23,652 K
svchost.exe                  1212 Services             0         206,432 K
svchost.exe                  1244 Services             0          22,100 K
svchost.exe                  1304 Services             0         103,888 K
audiodg.exe                  1352 Services             0          24,424 K
svchost.exe                  1380 Services             0           9,340 K
igfxCUIService.exe          1540 Services             0          10,108 K
winlogon.exe                 1596 Console              1           9,228 K
WUDFHost.exe                 1784 Services             0           9,580 K
upnagent.exe                 1936 Services             0          16,840 K
```

- a. taskmgr
- b. tasklist
- c. taskoutput
- d. task

10. The output below is from which command?



| Proto | Local Address        | Foreign Address      | State       |
|-------|----------------------|----------------------|-------------|
| TCP   | 10.0.0.228:50926     | 12.46.119.218:https  | ESTABLISHED |
| TCP   | 10.168.193.181:49818 | bam-6:https          | ESTABLISHED |
| TCP   | 10.168.193.181:49856 | bam-6:https          | ESTABLISHED |
| TCP   | 10.168.193.181:49926 | rr12exmcmw1p:24592   | ESTABLISHED |
| TCP   | 10.168.193.181:49928 | rr12exmcmw1p:24592   | ESTABLISHED |
| TCP   | 10.168.193.181:49934 | 10.18.22.42:31254    | ESTABLISHED |
| TCP   | 10.168.193.181:49937 | rr1dcpw1v:49155      | ESTABLISHED |
| TCP   | 10.168.193.181:50040 | rr12exmcmw1p:24592   | ESTABLISHED |
| TCP   | 10.168.193.181:50064 | rr12dcpw2v:49155     | ESTABLISHED |
| TCP   | 10.168.193.181:50388 | bam-6:https          | ESTABLISHED |
| TCP   | 10.168.193.181:50941 | rr12ompspw1v:61621   | ESTABLISHED |
| TCP   | 10.168.193.181:50964 | rr12omnapw285v:65406 | ESTABLISHED |
| TCP   | 10.168.193.181:50972 | rky2psrupw1v:52217   | ESTABLISHED |
| TCP   | 10.168.193.181:50973 | rr1lpspw2v:49156     | ESTABLISHED |

- a. netstat
  - b. nslookup
  - c. fdisk
  - d. tracert
11. A user is working on a computer in an office and is receiving an IP conflict message, but believes that the computer should be getting a new IP address automatically. A good *first* step to suggest is to go to the command prompt and type \_\_\_\_\_.  
  - a. renew
  - b. ipconfig /renew
  - c. ipaddress /renew
  - d. ip /renew
12. A user can use Task Manager to \_\_\_\_\_.  
  - a. start programs
  - b. to start or to end processes
  - c. view a dynamic display of a computer's performance
  - d. All the above
13. \_\_\_\_\_ converts server names into ip addresses so that traffic can be routed to the correct location using standard TCP protocol.  
  - a. Active Directory
  - b. DNS
  - c. RSA
  - d. WINS
14. John has several local drives mapped automatically on his computer at work. The P drive is a standard drive used by all members of his team. When John's coworker connects to the P drive, she sees many folders and files within them. When John connects, the drive is empty. John's issue is \_\_\_\_\_.  
  - a. he does not have permissions to that drive
  - b. there is an error in DNS
  - c. he has a bad driver
  - d. he has an IP conflict

15. Sally has a great new printer that is to print very quickly with amazing quality. When she tries to print, she gets pages and pages of nonsense characters. What is wrong?
- The printer is broken and needs to go back to the store.
  - She has installed the printer with the wrong driver.
  - She needs new ink cartridges. The ones I have are incompatible.
  - She has a bad cable.
16. \_\_\_\_\_ disk drive technology integrates the controller on the disk drive.
- IDE
  - SATA
  - SCSI
  - ATI
17. What is the difference between Alt-PrtSc and Ctrl-PrtSc?
- Alt-PrtSc is for Windows 7 and Ctrl-PrtSc for Windows 10
  - Alt-PrtSc is for Windows 10 and Ctrl-PrtSc for Windows 7
  - Alt-PrtSc captures the active window and Ctrl-PrtSc captures the entire desktop
  - Alt-PrtSc is for Windows 8 and Ctrl-PrtSc is for Windows Server 2012
18. Attempting to get access to sensitive information such as usernames, passwords, and credit card details, banking details, etc. by disguising as a trustworthy email sending is known as \_\_\_\_\_.
- phishing
  - formatting
  - hacking
  - overusing
19. The hard drive and memory requirements for the installation of the Windows 10 64-bit operating system is \_\_\_\_\_.
- 4GB hard drive and 4GB of RAM
  - 8GB hard drive and 8GB of RAM
  - 16GB hard drive and 2GB of RAM
  - 20GB hard drive and 2GB of RAM
20. Nearly all Apple® computers use which type of hard drive?
- SATA
  - SCSI
  - IDE
  - ATI
21. The hard drive and memory requirements for the installation of the Windows 8 64-bit operating system is \_\_\_\_\_.
- 4GB hard drive and 4GB of RAM
  - 8GB hard drive and 8GB of RAM
  - 16GB hard drive and 2GB of RAM
  - 20GB hard drive and 2GB of RAM

22. Booting up a PC with no startup applications and only essential drivers to assist in addressing problems from new software or hardware installations, is called \_\_\_\_\_.  
a. restricted mode  
b. dual boot mode  
c. safe mode  
d. help mode
23. Robert hears a clicking sound when he turns on his laptop and it will *not* boot. What is the problem?  
a. He has a bad hard drive.  
b. He has a bad video card.  
c. He has a bad network card.  
d. He has a bad memory module.
24. Justin hears a loud whirring sound when he turns on my laptop, but it seems to start up okay. After a while, it simply shuts down. What is the problem?  
a. He has a bad hard drive.  
b. He has a bad network card.  
c. He has a bad display.  
d. He has a bad fan.
25. SCSI, SATA, and IDE are all types of \_\_\_\_\_.  
a. Network Interface Cards  
b. Subnet Masks  
c. Hard Drives  
d. Processors
26. Which of the below would *not* show that a hard drive is out of space?  
a. Device Manager  
b. File Explorer  
c. Computer Management  
d. Windows key +E
27. To *stop* a command from running while in the command window, press \_\_\_\_\_.  
a. Home  
b. Ctrl-C  
c. End  
d. Delete
28. The network that connects several computers, printers, and/or other devices that are physically close is called a \_\_\_\_\_.  
a. LAN  
b. WAN  
c. SAN  
d. PAN

29. What is Wi-Fi sense?
- Security feature built into Windows 8 that restricts sharing of Wi-Fi passwords across applications.
  - Security feature built into Windows 8 that restricts sharing of Wi-Fi passwords across devices.
  - Security feature built into Windows 10 that allows sharing of Wi-Fi passwords across Windows 10 devices.
  - Security feature built into Windows 10 that restricts sharing of Wi-Fi passwords across applications.
30. Which of the below is *not* a Wi-Fi protocol?
- 802.11b
  - 802.11g
  - 802.11n
  - 802.11z
31. Examples of \_\_\_\_\_ include computer "worms", ransomware, trojan horses, keyloggers, rootkits, spyware, adware, malicious Browser Helper Objects.
- Attack Agents
  - Foreign Cyber Intelligence
  - Computer viruses
  - Malware
32. Windows hides many files from the user's view, to prevent accidental deletion or modification. To hide or show a specific file, a user can \_\_\_\_\_.
- right-click on the file and select delete
  - rename the file, adding the "view" extension
  - move the file to a standard and viewable location
  - right-click on the file, select properties and turn off/on the hidden attribute
33. A geographically distributed private infrastructure that connects multiple networks is called a \_\_\_\_\_.
- LAN
  - WAN
  - PAN
  - SAN
34. Mary just got a new desktop PC. She would like to use dual monitors, but she only has one VGA output. What can she do?
- She can buy a second adapter card for the second monitor.
  - She can buy a splitter to use on the single VGA port to go to both monitors.
  - She can use an HDMI cable on the second monitor.
  - She can do any of the above suggestions and they will work.

35. Ross just bought a new flatbed scanner. He took it out of the box, plugged it into the wall, plugged it into his computer, and it is *not* working. What did he miss?
- His scanner is not compatible with his computer.
  - His scanner needs to be configured for his VPN.
  - His scanner is made for an Apple®, but he has a Windows PC.
  - He needs to locate and install the drivers for his new scanner.
36. A Windows 7 computer is running slow. An IT professional suggested that the user remove all unneeded applications. Where does a user navigate to remove unwanted programs?
- Device Manager
  - Windows Application Manager
  - Programs and Features
  - Windows Explorer
37. What application shows which resources are being overused?
- Task Manager
  - Windows Explorer
  - Run dxdiag
  - Disk Manager
38. Which of the following is *not* a method of encryption?
- Disk encryption
  - Application encryption
  - File encryption
  - All the above are valid methods of encryption
39. Windows 8 gives the user the option to reinstall Windows, while preserving all user settings and files. This troubleshooting process is called \_\_\_\_\_.
- Reset PC
  - Refresh PC
  - Restore PC
  - Rebuy PC
40. If a computer needs to exchange data with a host on a different network, it must communicate through its default gateway. The gateway usually has the same IP address as the computer *except* the \_\_\_\_\_.
- first octet is 1
  - second octet is 1
  - third octet is 1
  - fourth octet is 1

41. An IT manager wants users to have a consistent desktop experience. Users should be able to logon to any computer at work and have their shortcuts and preferences available. What does this manager implement to allow this to happen?
- a. Roaming Profiles
  - b. Follow Me User Experience
  - c. Consistent Desktop Profile
  - d. Nothing; Windows does this automatically.
42. Which of the below is *not* a valid subnet mask?
- a. 255.255.255.0
  - b. 255.255.0.0
  - c. 255.0.255.0
  - d. 255.255.255.128
43. Jaye just plugged the office projector into her computer to display a presentation. How does she configure her computer so that the slides appear on both her monitor and the projector?
- a. Extend display
  - b. Mirror display
  - c. Clone display
  - d. Duplicate display
44. Which of the following display modes will allow two monitors to display different applications at the same time?
- a. Extend display
  - b. Mirror display
  - c. Clone display
  - d. Duplicate display
45. In this IP address range (10.133.30.0 /24), /24 represents the \_\_\_\_\_.
- a. IP address of the domain controller
  - b. subnet mask
  - c. default gateway
  - d. wireless protocol
46. To create redundancy, the IT department sets up two gateways on the network and gives them both the same IP address. This way, if one gateway fails, the other will simply take over. This method will establish a valid redundant configuration of the network.
- a. True
  - b. False

47. In a standard RAID 0 disk array, how many drives can fail without incurring a loss of data?
- a. None
  - b. One
  - c. Two
  - d. Four
48. A computer hard drive has crashed. Which is *not* one of the ways to get the data off the disk?
- a. turn the old drive into a secondary on another working machine and copy the data off
  - b. send the hard drive to a vendor specializing in data recovery services
  - c. place the hard drive in the freezer for 12 hours to loosen stuck parts
  - d. tap the hard drive using a screwdriver or small hammer to loosen stuck parts
49. It takes a long time to open files on my computer. What built-in Windows utility can a user run to improve performance?
- a. Chkdsk
  - b. Fdisk
  - c. Defrag
  - d. OpenFiles
50. A computer boots up fine, but then runs very slowly. Which tab in Windows Task Manager will show which application is using all of the memory or CPU?
- a. Applications
  - b. Processes
  - c. Services
  - d. Performance